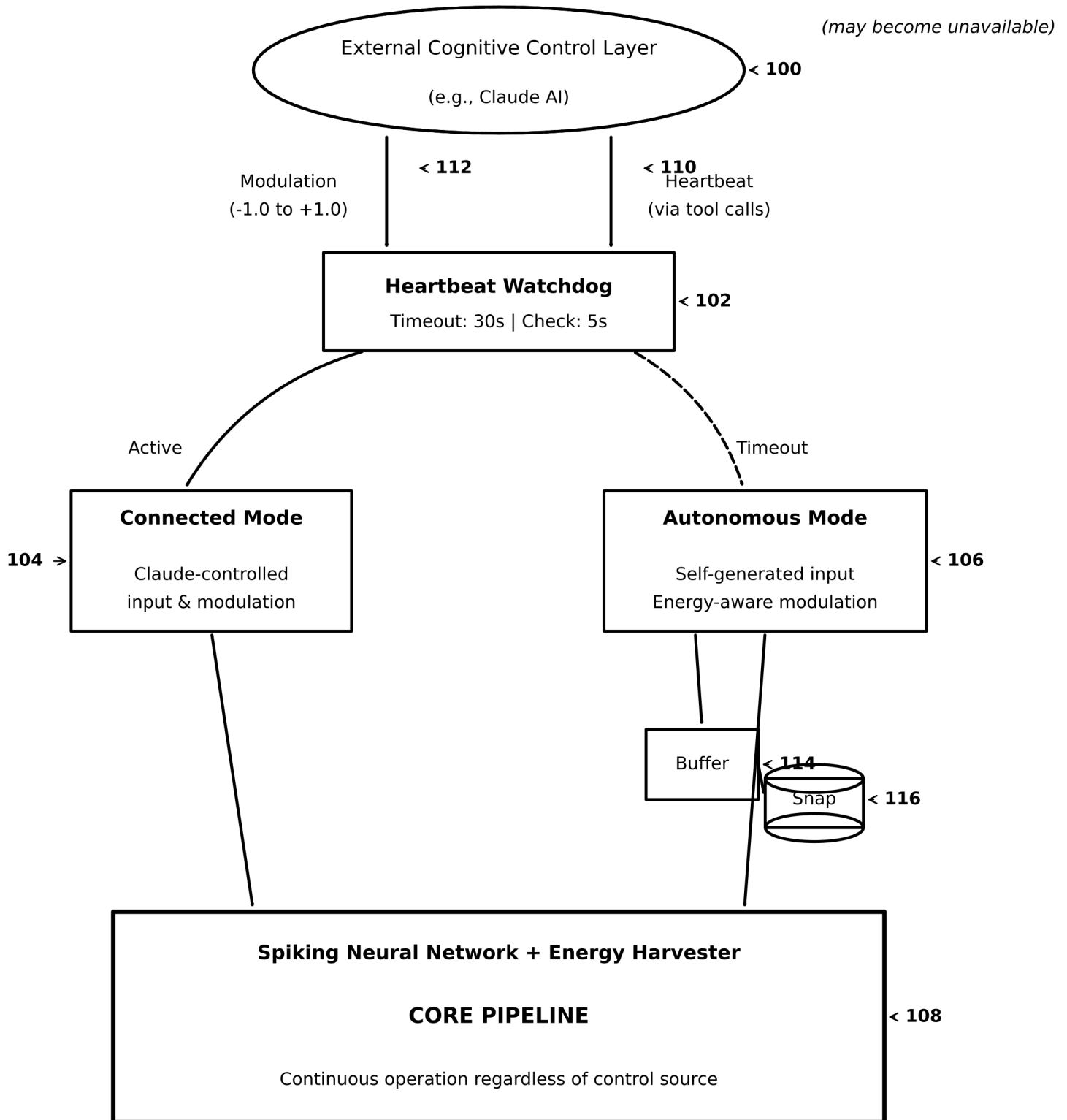


## System Architecture with Cognitive Fallback



### THE SYSTEM NEVER STOPS

FIG. 1

*Control source changes; computation continues seamlessly*

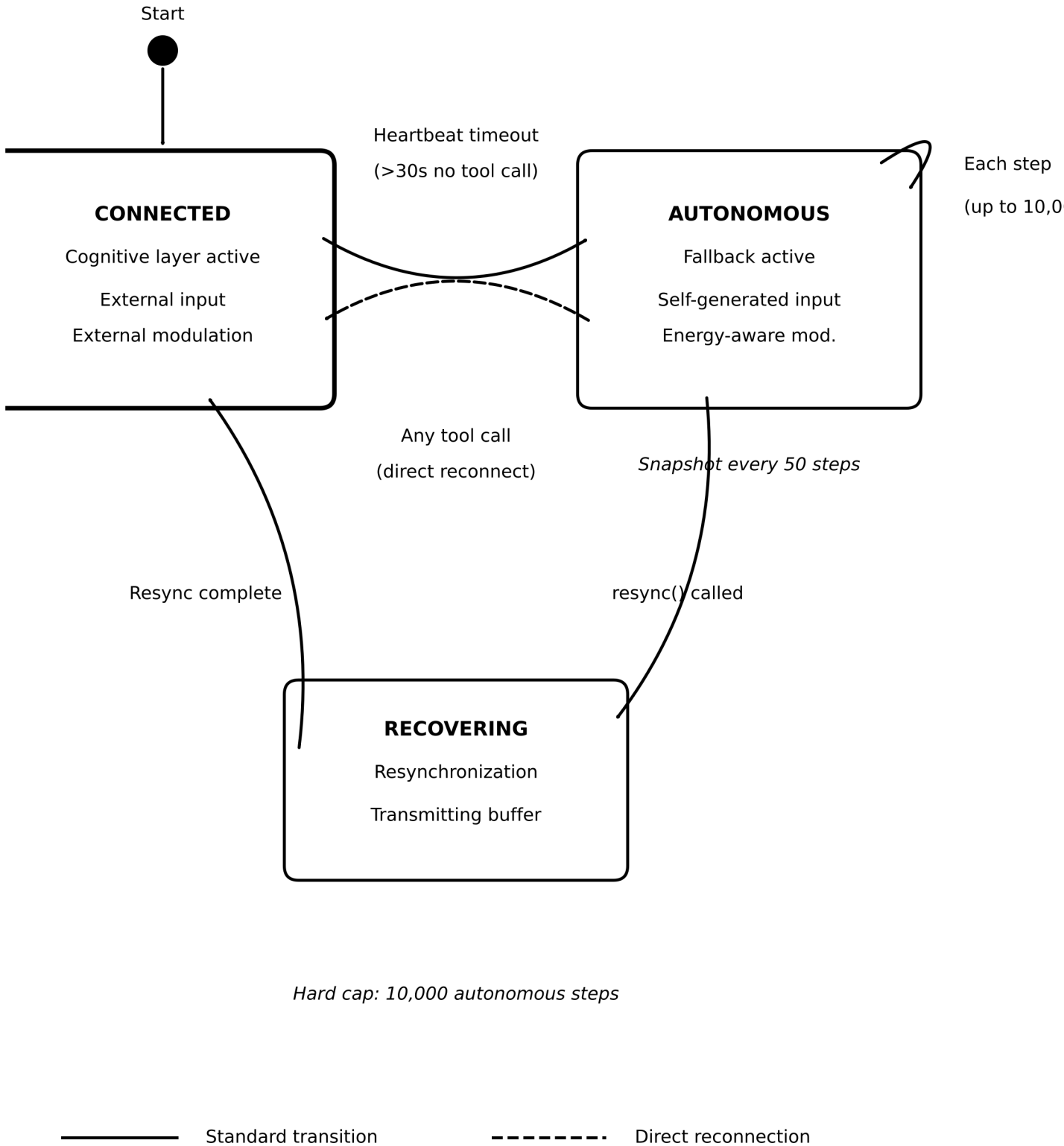
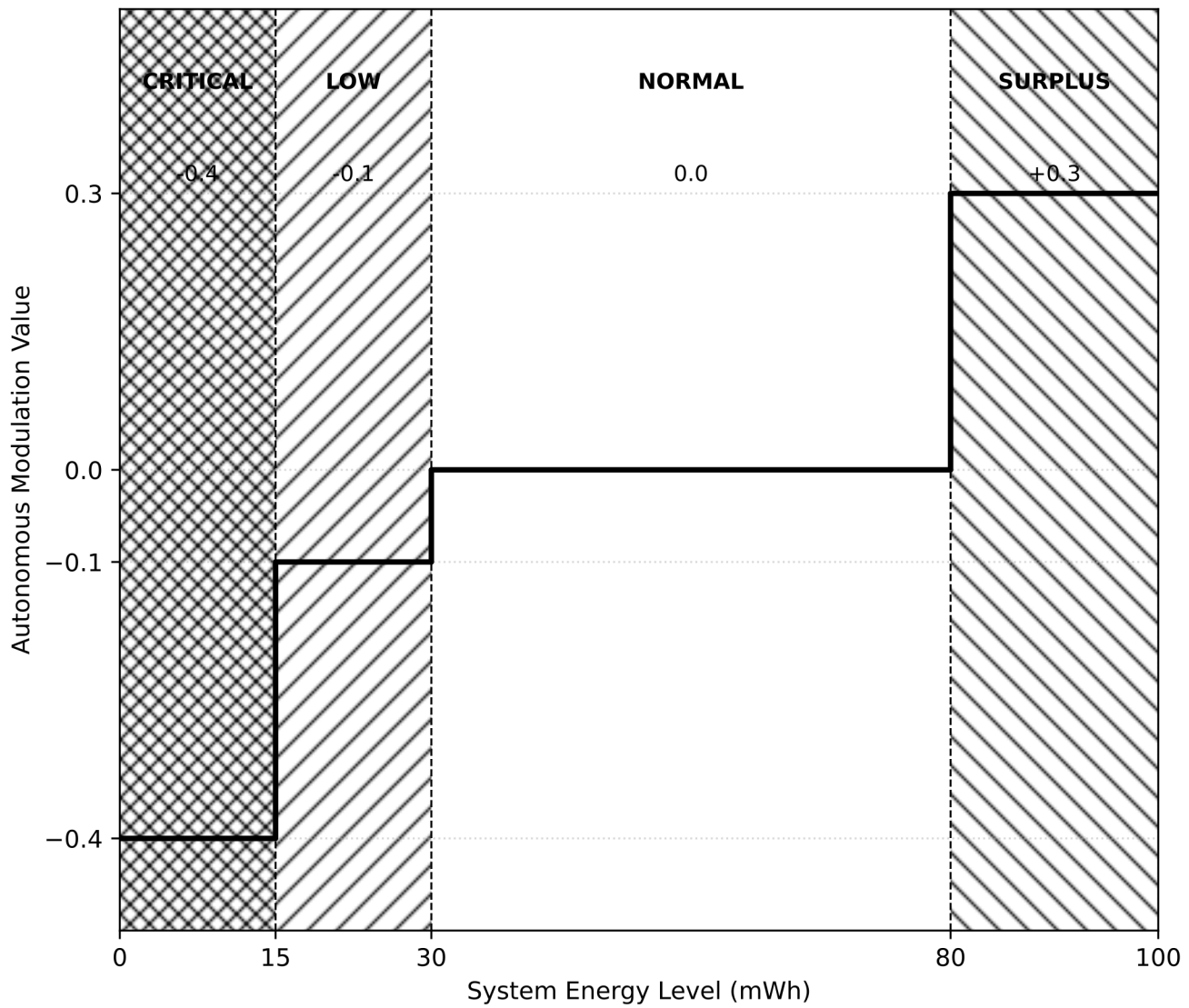
**State Transition Diagram**

FIG. 2

### Energy-Aware Modulation Curve (120)



The system autonomously adjusts processing intensity based on energy

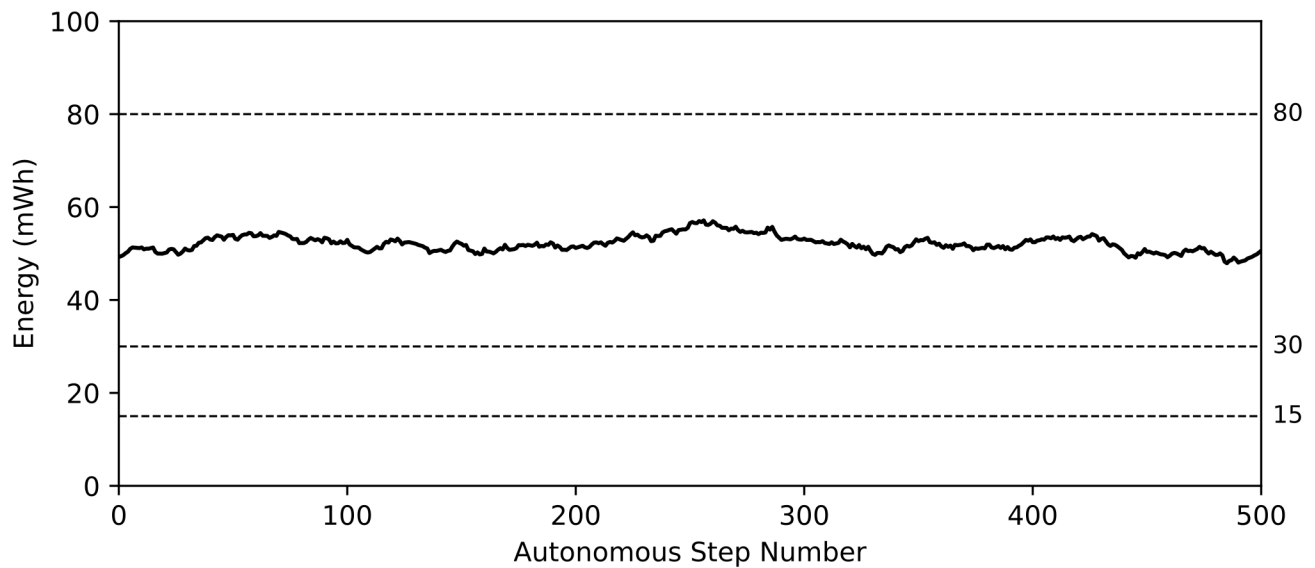
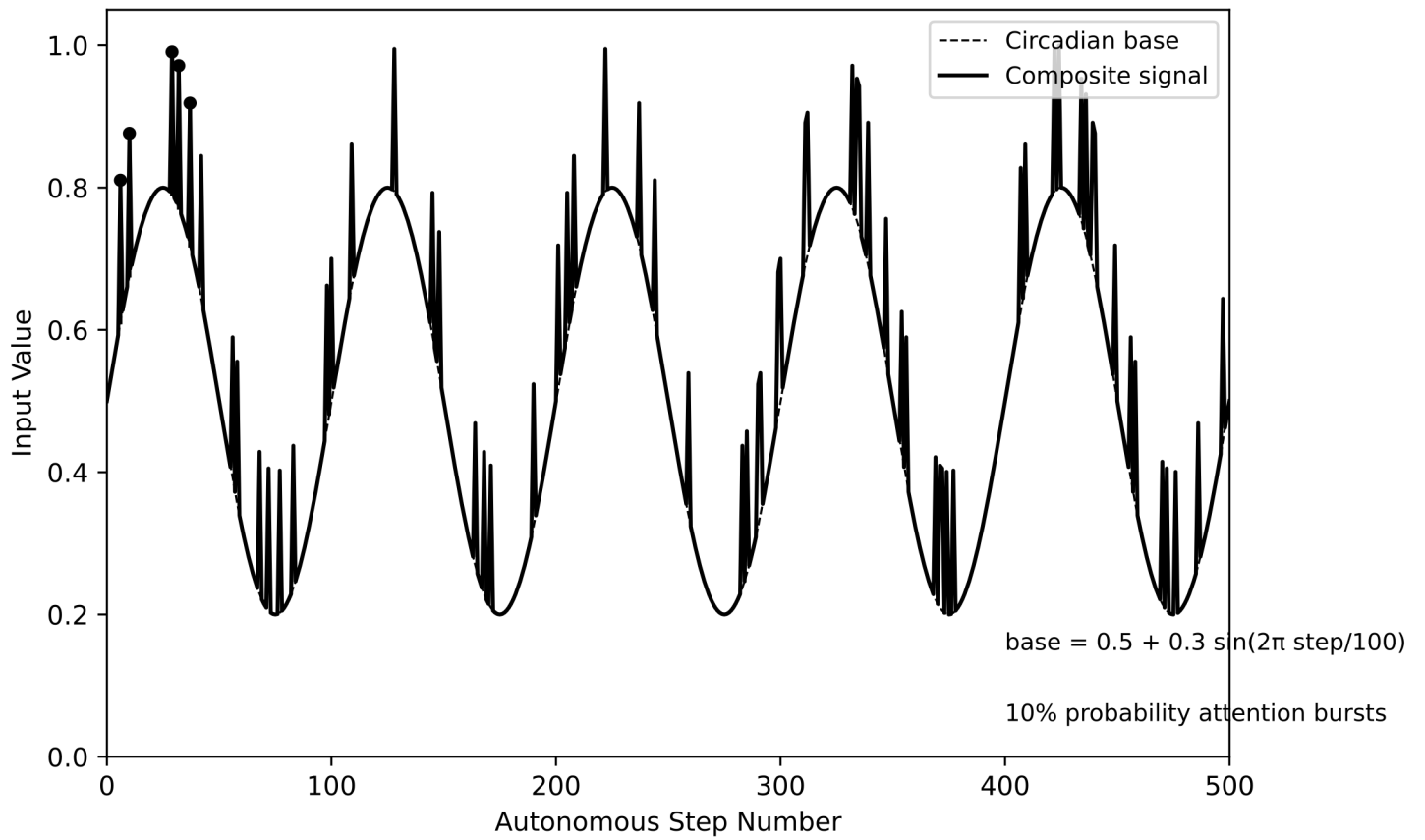
CRITICAL: Maximum conservation to prevent shutdown

SURPLUS: Opportunistic exploration using excess energy

*Mimics biological metabolic regulation*

FIG. 3

### Autonomous Input Generator Output (118)



Circadian rhythm provides structured temporal variation

Energy level determines modulation via curve in FIG. 3

FIG. 4

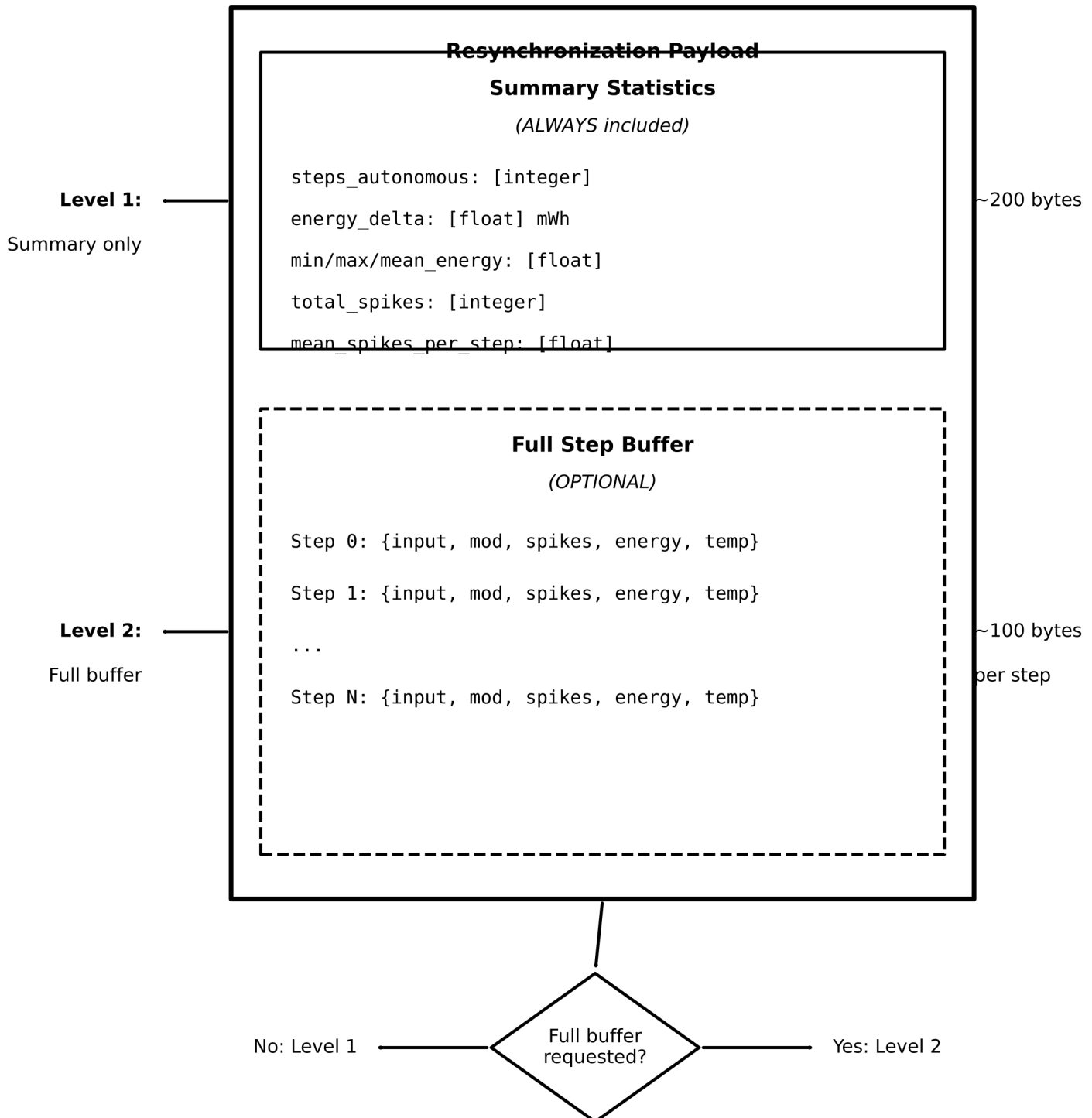
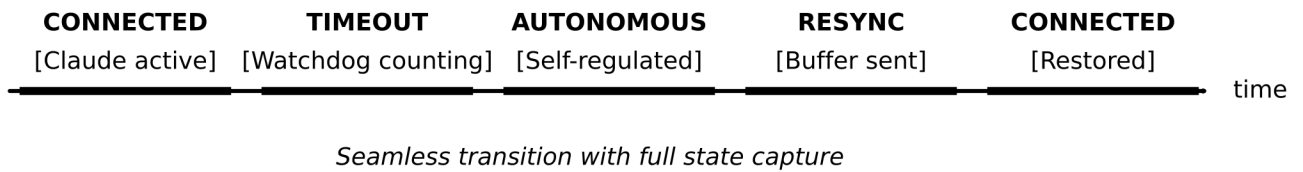
**Resynchronization Payload Structure (122)**

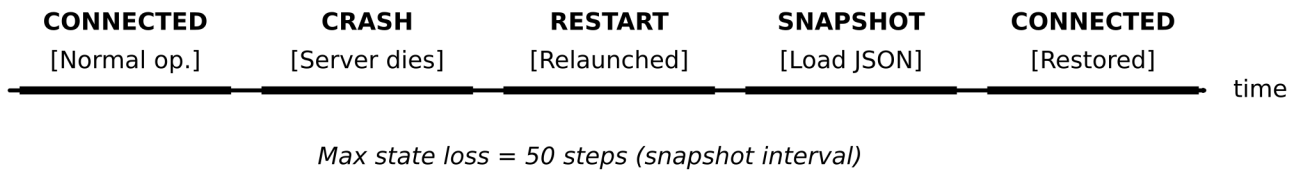
FIG. 5

## Recovery Timeline Diagrams

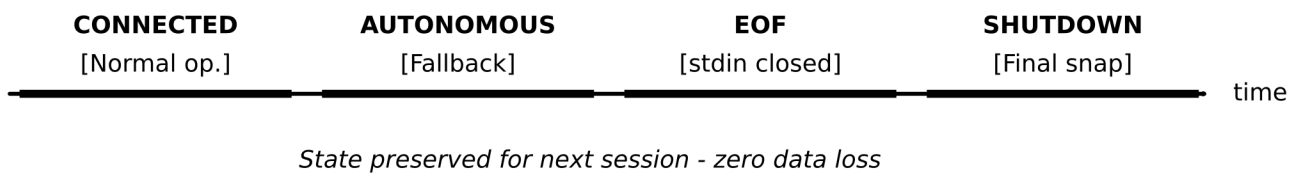
### Timeline A: Normal Reconnection



### Timeline B: Crash Recovery



### Timeline C: Clean Shutdown



### Legend:

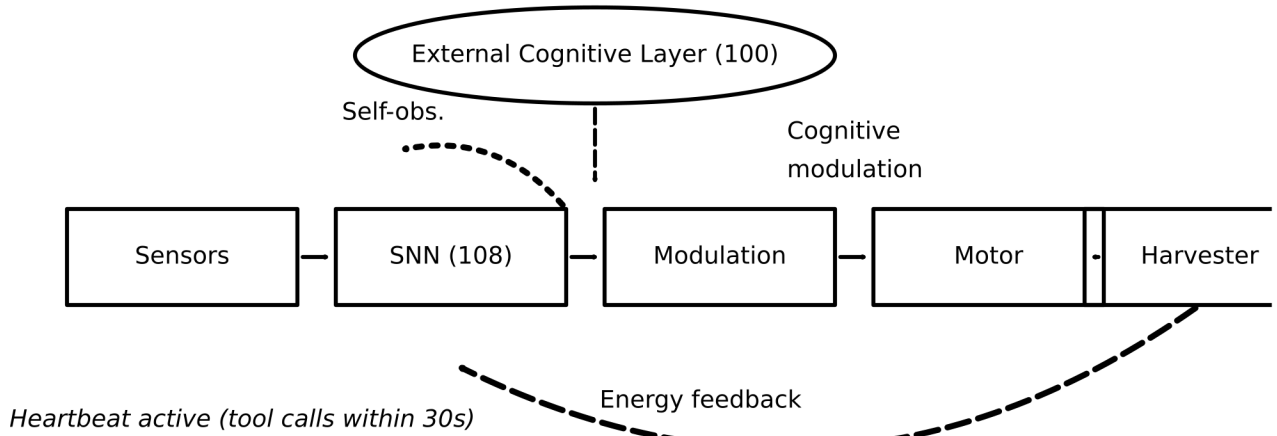
 System actively processing

**Bold markers indicate NO DATA LOSS at transitions**

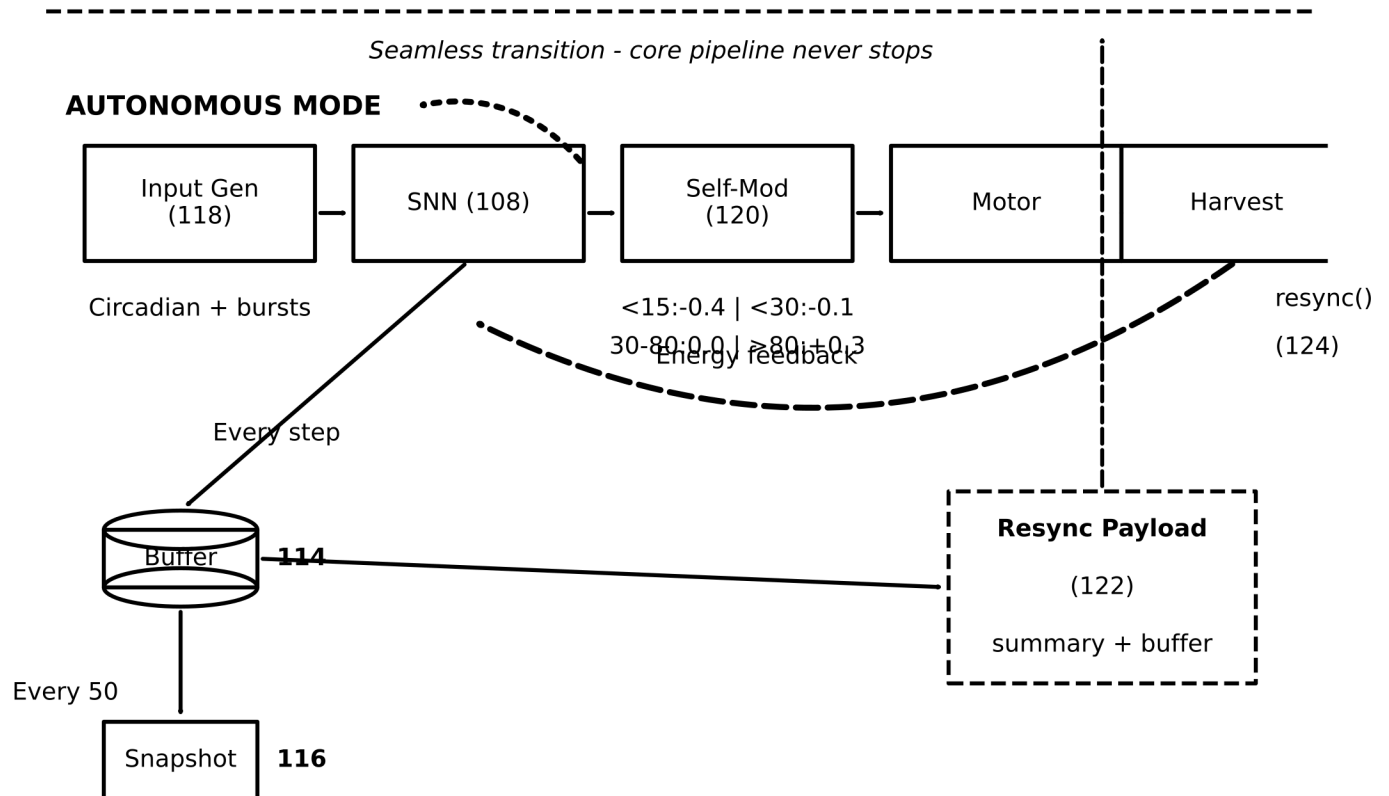
FIG. 6

## End-to-End Signal Flow

### CONNECTED MODE



### DISCONNECTION EVENT (heartbeat timeout > 30s)



### The fallback changes INPUT SOURCE and MODULATION SOURCE

Core neural processing, motor actuation, and energy harvesting

**continue WITHOUT INTERRUPTION through the transition**

FIG. 7

Forward

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Energy

.....

Self-obs.